

MAXILLOFACIAL INJURIES

SECURE THE AIRWAY FIRST THEN STOP TORRENTIAL HAEMORRHAGE. MOST PATIENTS CAN BE CONVEYED SITTING UP TO AVOID AIRWAY COMPLICATIONS.

ASSESS INJURY and CONSIDER EARLY REQUEST FOR ADVANCED CARE IF AVAILABLE⁽¹⁾

MINOR / MODERATE MAXILLOFACIAL INJURY
ALERT AND ABLE TO OBEY COMMANDS
ABLE TO MAINTAIN OWN AIRWAY BY SITTING UP
AND
MINOR OR COMPRESSIBLE BLEEDING

MAINTAIN SUITABLE POSITION (SITTING UP OR POSTURAL DRAINAGE) TO PROTECT THE AIRWAY

FULL MONITORING, BREATHING, CIRCULATION AND HEAD INJURY ASSESSMENT

IV ACCESS AND GIVE ADEQUATE ANALGESIA
CONSIDER TXA WHILST CONVEYING

IF BLEEDING INTO ORAL CAVITY SOAK GAUZE IN 50/50 SOLUTION OF TXA AND SALINE AND ASK THE PATIENT TO BITE GAUZE AND LEAN FORWARD

CONVEY TO NEAREST MTC OR MEDICAL FACILITY WITH SURGICAL CAPABILITY
MONITOR FOR DETERIORATION ON ROUTE

SIGNIFICANT MAXILLOFACIAL INJURY⁽²⁾
REDUCED CONSCIOUS LEVELS, SEVERE HEAD INJURY OR UNABLE TO MAINTAIN AIRWAY AND
TORRENTIAL HAEMORRHAGE⁽³⁾

MAINTAIN SUITABLE POSITION TO PROTECT AIRWAY AND ATTEMPT PRE-OXYGENATION

PREPARE DEFINITIVE AIRWAY KIT DUMP & SUCTION
DISCUSS AIRWAY PLAN - SURGICAL OR INTUBATE?

GAIN BILATERAL IV ACCESS FOR SEDATION AND PREHOSPITAL BLOOD PRODUCTS IF REQUIRED

A

ONCE KIT DUMP AND AIRWAY CLINICIAN IS READY: PLACE PATIENT IN POSITION TO MANAGE AIRWAY
ASSISTANT PROVIDES CONTINUOUS SUCTION OR AIRWAY CLINICIAN USES SALAD TECHNIQUE⁽⁴⁾
PERFORM PLANNED AIRWAY PROCEDURE

TAMPONADE BLEEDING WITH EPISTATS

TAMPONADE TORRENTIAL HAEMORRHAGE WITH EPISTATS

1. ENSURE DEFINITIVE AIRWAY IS SECURE FIRST (SEE ABOVE OR ADVANCED AIRWAY GUIDE).
2. INSERT DENTAL BRIDGES AND POSITION BETWEEN MOLARS WITH THE POINT OF WEDGE TOWARD THE BACK OF THE MOUTH AND SECURE WITH CHAIN OR TAPING TO CHEEK.
3. APPROPRIATE SIZE AND FIT A CERVICAL COLLAR TO SECURE THE MANDIBLE IN PLACE.
4. INSERT BOTH EPISTATS ALONG THE FLOOR OF THE NOSE (SIMILAR TO NPA INSERTION).
5. USING 20ml SYRINGES OF SALINE, SLOWLY INFLATE THE BALLOONS IN THE POSTERIOR NASAL SPACE (WHITE VALVE) AT THE SAME TIME WITH 5ml ALLIQUOTS TO A MAXIMUM OF 10ml EACH.
6. NOW INFLATE THE MIDDLE BALLOONS (CLEAR VALVE) OF EACH EPISTAT TOGETHER WITH 5ml AT A TIME UNTIL HAEMORRHAGE IS CONTROLLED – TYPICALLY 20-30ml.

ONGOING MANAGEMENT AND TRANSFER

AIRWAY	ENSURE AIRWAY IS SECURE AND ETCO ₂ HAS GOOD WAVEFORM UNSTABLE MANDIBULAR FRACTURES WILL REQUIRE AIRWAY SUPPORT BY EITHER POSITIONING, PATIENT HOLDING THEIR OWN JAW, MAINTAINING JAW THRUST IN REDUCED CONSCIOUSNESS OR ADVANCED AIRWAY MANAGEMENT.
BREATHING	MAINTAIN SpO ₂ >95%. CONSIDER MISSING TEETH MAY HAVE BEEN ASPIRATED.
CIRCULATION	CONTINUOUS HR AND BP MONITORING ASSESSING FOR SIGNS OF SHOCK. BLOOD OR FLUID RESUSCITATION WILL LIKELY BE REQUIRED FOR TORRENTIAL HAEMORRHAGE PATIENTS OR SEVERE HEAD INJURED PATIENTS.
DISABILITY/ HEAD	THOROUGH HEAD INJURY ASSESSMENT ⁽⁵⁾ (GCS AND PUPILS) FOR DETERIORATION IN PATIENTS WITH MINOR OR MODERATE MAXILLOFACIAL INJURIES. ASSESS FOR SIGNS OF WORSENING INTRACRANIAL HERNIATION IN SEVERELY HEAD INJURED PATIENTS (CUSHING’S RESPONSE OR DILATED PUPILS) AND CONSIDER IV HYPERTONIC SALINE 5% (6ml/kg MAX 350ml)

POINTS TO NOTE

1. EARLY REQUEST IS NEEDED FOR LEVEL 8 OR CRITICAL CARE SUPPORT IN SIGNIFICANT INJURY AS THE PATIENT IS LIKELY TO REQUIRE PREHOSPITAL EMERGENCY ANAESTHESIA FOR DEFINITIVE AIRWAY MANAGEMENT. IF THIS IS UNAVAILABLE, EMERGENCY CONVEYANCE TO THE NEAREST SURGICAL CAPABILITY IS REQUIRED, MAINTAINING POSTURAL DRAINAGE OR CONSIDER SURGICAL AIRWAY IF TRAINED TO DO SO IN HAEMORRHAGING PATIENTS.
2. 10% OF SIGNIFICANT MAXILLOFACIAL FRACTURES HAVE TORRENTIAL HAEMORRHAGE, MAINLY FROM THE MAXILLARY ARTERY. MANUAL REDUCTION OF FRACTURES CAN PROVIDE SOME HAEMOSTASIS BUT DEFINITIVE AIRWAY MANAGEMENT AND BALLOON TAMPONADE IS BEST PRACTICE IN THESE PATIENTS (Jose *et al.*, 2016).
3. DEFINITIVE AIRWAY CONTROL MUST BE PERFORMED (CONSIDER SEDATION PRIOR) BEFORE TAMPONADE OF TORRENTIAL HAEMORRHAGE CAN BEGIN.
4. SALAD (SUCTION ASSISTED LARYNGOSCOPY AIRWAY DECONTAMINATION) INVOLVES THE AIRWAY CLINICIAN USING THE SUCTION HARD CATHETER TO FACILITATE LARYNGOSCOPY OF THE SOILED AIRWAY. CONTINUOUS SUCTIONING SHOULD BE PERFORMED BY THE AIRWAY ASSISTANT SO THE AIRWAY CLINICIAN CAN PERFORM INTUBATION OR SURGICAL AIRWAY.
5. PATIENTS SUFFERING FROM SEVERE MAXILLOFACIAL INJURIES ARE PRESUMED TO HAVE A HEAD INJURY UNTIL PROVEN OTHERWISE (REFER TO HEAD INJURY MANAGEMENT GUIDANCE).
6. MAINTAINING MANUAL IN-LINE STABILISATION IS IMPORTANT SIGNIFICANT INJURY PATIENTS BUT AIRWAY REMAINS A PRIORITY OVER C-SPINE INJURIES, SO CONVEY IN A POSITION THAT ALLOWS THE AIRWAY TO BE MAINTAINED AND ASSESS FOR NEUROLOGICAL FUNCTION ON ROUTE.
7. ANY MISSING TEETH THAT HAVE BEEN FOUND SHOULD BE GENTLY CLEANED AND REINSERTED INTO THE GUM SOCKET THEN ENCOURAGED TO BITE ON GAUZE TO KEEP TOOTH IN LOCATION IF FEASIBLE.
8. UNSTABLE MANDIBULAR FRACTURES CAN CAUSE AIRWAY OBSTRUCTION, ESPECIALLY IF THE MANDIBLE IS NOT SUPPORTED. IF ALERT AND COHERENT, GET PATIENT TO LEAN FORWARD AND SUPPORT THEIR OWN MANDIBLE AND TRANSFER TO NEAREST TRAUMA UNIT. IF THE PATIENT HAS REDUCED CONSCIOUSNESS BUT NO HAEMORRHAGE, THEN PROCEED WITH JAW THRUST OR BASIC AIRWAY MANOEUVRES BUT PREPARE FOR DETERIORATION AND NEED FOR ADVANCED AIRWAY MANAGEMENT.
9. HYPOXIA AND HYPOVOLAEMIA ARE MOST LIKELY CAUSES OF CARDIAC ARREST SO MANUAL VENTILATION AND FLUID RESUSCITATION WILL BE REQUIRED. TREAT AS TRAUMATIC CARDIAC ARREST.

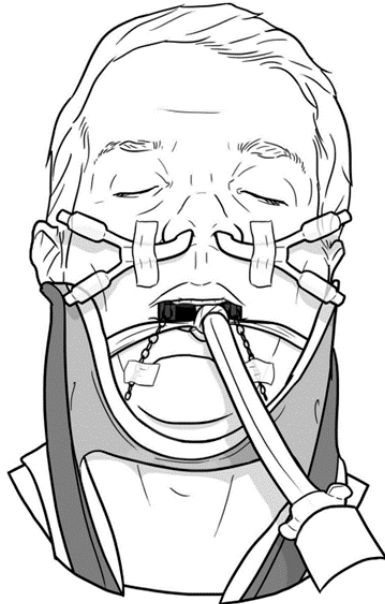


Diagram of EpiStat and Maxillofacial Packing with Endotracheal Intubation

References

Guideline written in accordance with SFM Clinical Standard Operating Guideline 016 'Maxillofacial Injuries' (2022) and MERT Standard Operating Procedure 'Maxillofacial Haemorrhage V3.0'.

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